

Explainable Artificial Intelligence in PDAC diagnosis

This survey will show you some outputs from two explainable AI techniques: SHAP and LIME. Most questions will ask you to indicate whether you agree or disagree with the values and results emerging from these techniques, using a scale from 1 to 5.

The purpose of this survey is to collect qualitative evaluation data for my master thesis at WU Vienna, focusing on a multimodal Explainable AI evaluation in the context of PDAC diagnosis.

The results will be used to evaluate these AI based techniques on a qualitative level and foster the adoption of non interpretable models, like neural networks, in sensitive domains like PDAC/cancer diagnosis.

This survey intends to remain anonymous, for any inquiries or particular acknowledgment needs, do not hesitate to reach out to ariana.calderara@gs.wu.ac.at

Thank you in advance for your precious contribution!

— * Indica una domanda obbligatoria

Before you begin, key terms explained

This survey uses a small number of technical terms. You do not need any background in artificial intelligence to participate. The definitions below are all you need to understand what you are looking at

PDAC (Pancreatic Ductal Adenocarcinoma): the most common and aggressive form of pancreatic cancer. It is often diagnosed at a late stage, which is why early detection using clinical biomarkers is an active area of research.

Biomarker: a measurable biological indicator found in blood, urine, or tissue that can signal the presence or progression of a disease. In this study, the biomarkers used include urinary and blood-based proteins such as CA19-9, REG1A, REG1B, TFF1, and LYVE1, which have been studied in the context of PDAC detection.

AI/Machine Learning Model: computer program trained on patient data to recognize patterns and make predictions, in this case, whether a patient is likely to be Healthy, have a Benign condition, or have PDAC (Cancer). The models used in this study are Random Forest, Support Vector Machine, and Neura Network. You do not need to understand how these work internally; **what matters here is whether their explanations make sense to you.**

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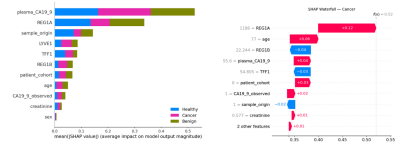
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4. What risks do you associate with AI systems employed in the PDAC/oncology domain?

Clinical Correctness: SHAP

In the left figure, the chart shows which biomarkers most influenced the AI model's predictions across all patients in the dataset. Each bar represents one biomarker. Longer bars mean the biomarker had a stronger average influence on the model's output. The three colors correspond to the three diagnostic classes: blue = Healthy, pink = Cancer, olive = Benign.

In the right figure, the chart explains why the model predicted **cancer** for one specific patient. Starting from the average baseline prediction (bottom of the chart), each bar shows how a particular biomarker pushed the prediction higher (red, towards cancer) or lower (blue, away from cancer).



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!! XAI – Explainable Artificial Intelligence: a set of techniques designed to make AI predictions more transparent and interpretable. Rather than just saying "this patient has cancer", an XAI technique also explains *why* – for example, by showing which biomarkers influenced the prediction and in which direction. This survey evaluates two such techniques: SHAP and LIME.

SHAP (SHapley Additive exPlanations): a technique that assigns each biomarker a score indicating how much it contributed to a specific prediction. A positive score means the biomarker pushed the prediction towards a given diagnosis; a negative score means it pushed it away. SHAP can show both a **global view** (which features matter most across all patients) and a **local view** (which features mattered for one specific patient). In the charts you will see, red bars push the prediction higher and blue bars push it lower.

LIME (Local Interpretable Model-agnostic Explanations): a technique that explains one individual prediction at a time by testing small variations in the patient's data and observing how the model's output changes. The result is a ranked list of conditions – for example, "REG1A > 551" – showing which feature ranges supported or opposed the prediction for that specific patient. In the charts you will see, green bars support the predicted diagnosis and red bars oppose it.

The three diagnostic classes Throughout this survey, predictions fall into one of three categories:

- **Healthy:** no signs of disease
- **Cancer:** PDAC diagnosis
- **Benign:** a non-cancerous pancreatic condition

All questions use a scale from 1 to 5, where:

- 1 = Strongly disagree
- 3 = Neutral
- 5 = Strongly agree

There are no right or wrong answers. Your clinical or scientific intuition is exactly what this survey is designed to capture.

Background

The following questions are asked for segmentation purposes

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5. The attributes highlighted as the most important seem relevant to PDAC diagnosis *

Contrassegna solo un ovale.

6. The explanation is consistent with what I would expect clinically or scientifically *

Contrassegna solo un ovale.

7. Based on this explanation, I understand why the model made this prediction *

Contrassegna solo un ovale.

8. I trust this prediction given the explanation provided *

Contrassegna solo un ovale.

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1. What best describes your role? *

Contrassegna solo un ovale.

2. How many years of experience do you have working with PDAC? *

Contrassegna solo un ovale.

3. How familiar are you with AI and machine learning technologies? *

Contrassegna solo un ovale.

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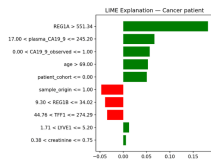
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9. The attributes highlighted as the most important seem relevant to PDAC diagnosis *

Contrassegna solo un ovale.

Clinical Correctness: LIME

Green bars indicate features that supported the cancer prediction; red bars indicate features that worked against it. Each feature is shown as a value range (e.g. "REG1A > 551") rather than a single number.



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10. The explanation is consistent with what I would expect clinically or scientifically *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

11. Based on this explanation, I understand why the model made this prediction *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

12. I trust this prediction given the explanation provided *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

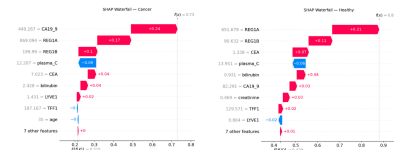
Clinical Consistency: SHAP

These two charts show SHAP explanations for two individual patients from the same dataset, one predicted as **Cancer** (figure on the left) and one predicted as **Healthy** (figure on the right). Both were generated by the same model.

In this section you are asked to evaluate **consistency**: do the explanations behave in a stable and predictable way across different patients?

For example, do the same biomarkers tend to appear as important drivers regardless of which patient is being examined?

A clinically consistent explanation should highlight features that align with established medical knowledge about PDAC, not shift arbitrarily between patients.



13. The two explanations highlight similar features for these similar cases

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

14. If I were shown these explanations on different days, I would expect the same result

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

15. I would feel confident relying on this explanation repeatedly in practice

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

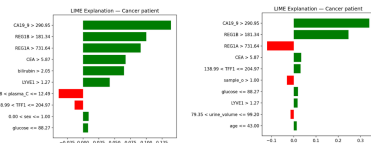
Clinical Consistency - LIME

Both charts show a LIME explanation for a **cancer prediction** on the same dataset, but generated by two different model architectures: figure on the right uses a **Random Forest**, figure on the left uses a **Neural Network (MLP)**.

The underlying patient data is the same class (cancer), but the models may identify different biomarkers as the main drivers.

In this section you are asked whether the LIME explanations feel **consistent** and **reliable**: if you received one of these explanations today and the other tomorrow for a similar patient, would you expect them to tell the same clinical story?

Inconsistency between models on the same class may signal that the explanation is sensitive to internal model choices rather than the patient's actual biology.



16. The two explanations highlight similar features for these similar cases *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

17. If I were shown these explanations on different days, I would expect the same result *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

18. I would feel confident relying on this explanation repeatedly in practice *

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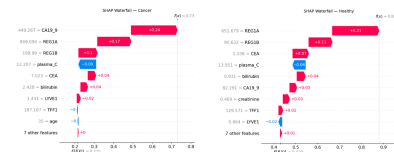
1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

Clinical Contrastivity: SHAP

These two charts show SHAP explanations for two patients with **clearly different diagnoses**: one predicted as Cancer and one predicted as Healthy, using the same model and dataset.

In this section you are asked to evaluate **contrastivity**: does the explanation visibly and meaningfully change when the patient's clinical picture changes?

A good explanation should look clearly different for a cancer patient versus a healthy one, reflecting the actual biological differences between these cases. If the two charts look nearly identical in which features are highlighted and in which direction, that would be a concern.



19. The explanation clearly reflects the differences between these two patients. *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

20. When patient data changes significantly, I would expect the explanation to change accordingly

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

21. This explanation would help distinguish between different patient risk profiles *

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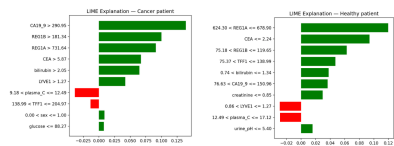
1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

Clinical Contrastivity: LIME

Similar to previous plots these two LIME charts explain predictions for a **cancer patient** and a **healthy patient**, using the same Random Forest model on the CPD dataset.

As with the SHAP contrastivity section, you are asked whether the explanation **reacts meaningfully** to the **difference** between these two patients.

Do the highlighted biomarker conditions and their directions (supporting vs opposing the prediction) clearly reflect what you would expect clinically for a cancer versus a healthy case?



22. The explanation clearly reflects the differences between these two patients. *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

23. When patient data changes significantly, I would expect the explanation to change accordingly

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

24. This explanation would help distinguish between different patient risk profiles *

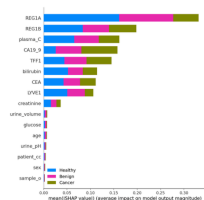
Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

Overall Clinical Transparency

This final chart provides an overall summary of which biomarkers drove the model's predictions across the full dataset, broken down by diagnostic class.

It is included here as a reference to support your overall impressions. Having now seen both SHAP and LIME explanations across multiple patients and contexts, the following questions ask for your general assessment of how understandable, trustworthy, and clinically useful these AI explanations are as a whole.



25. Overall, the proposed explanations are understandable given my background *

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

26. I would trust these explanations enough to use it in my work (clinical, research, or educational context)

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

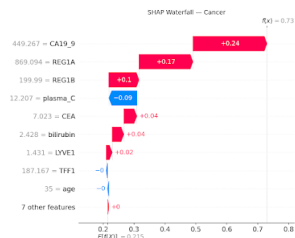
27. These explanations would be useful for communicating AI predictions to others in my field

Contrassegna solo un ovale.

1 2 3 4 5
Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

28. Based on everything you have seen in this survey, how would you rate the overall transparency of the SHAP explanations for clinical use? Please select the level that best matches your impression.

The fire below serves as an example for a SHAP explanation

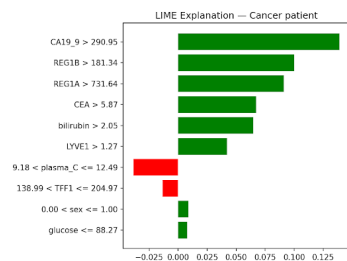


Contrassegna solo un ovale.

- ☐ The system provides no explanation at all
- ☐ Some explanation exists but it is not understandable to me
- ☐ I can partially understand the explanation but would not rely on it
- ☐ The explanation is understandable and I could follow the reasoning
- ☐ The explanation is clear, clinically meaningful, and I would trust it in practice
- ☐ The explanation is fully transparent and suitable for autonomous clinical use

29. Based on everything you have seen in this survey, how would you rate the overall transparency of the LIME explanations for clinical use? Please select the level that best matches your impression.

The fire below serves as an example for a LIME explanation



Contrassegna solo un ovale.

- ☐ The system provides no explanation at all
- ☐ Some explanation exists but it is not understandable to me
- ☐ I can partially understand the explanation but would not rely on it
- ☐ The explanation is understandable and I could follow the reasoning
- ☐ The explanation is clear, clinically meaningful, and I would trust it in practice
- ☐ The explanation is fully transparent and suitable for autonomous clinical use

30. Overall, which explanation technique did you find more useful for understanding the AI's predictions?

Contrassegna solo un ovale.

- ☐ SHAP (the bar charts showing feature importance scores)
- ☐ LIME (the charts showing feature value ranges)
- ☐ Both equally useful
- ☐ Neither was useful
- ☐ I cannot tell the difference

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Google Moduli

